



Introduction

Agglomerations and the rise of urban network externalities*

Martijn J. Burger¹, Evert J. Meijers²

¹ Department of Applied Economics, Erasmus University, Rotterdam, Tinbergen Institute and academic director at the Erasmus Happiness Economics Research Organization, PO Box 1738, 3000 DR Rotterdam, The Netherlands (e-mail: mburger@ese.eur.nl)

² Faculty of Architecture and the Built Environment, Delft University of Technology, Julianalaan 134, 2628 BL Delft, The Netherlands (e-mail: e.j.meijers@tudelft.nl)

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Abstract. This paper introduces the theme of the special issue ‘Agglomerations and the Rise of Urban Network Externalities’. Urban network externalities are defined as external economies from which firms and households can benefit by being located in agglomerations that are well embedded in networks that connect with other agglomerations. The contributions focus on the conceptualization of urban network externalities and their influence on urban performance. Finally, a research agenda is presented, that should focus on multiplexity and heterogeneity in networks and their impacts; interrelations between agglomerations and networks and their dynamic and place-based nature; and, the policy implications of urban network externalities.

JEL classification: O18, R11, R12

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1 Introduction to the special issue

Urban and regional planners, geographers and economists alike are interested in the forces that create, shape and maintain concentrations of economic activity. A large body of empirical literature has emerged that addresses economic geography, regional science, and urban economics and investigates how spatial circumstances give rise to agglomeration externalities – external economies from which firms can benefit through co-location (e.g., Glaeser et al. 1992; Henderson et al. 1995; Combes 2000; Rosenthal and Strange 2003).

In most of the early empirical studies, agglomeration externalities are defined as being geographically constrained, and it is assumed that economic activities outside a certain area have no effect on the economic activities within that area (Rosenthal and Strange 2004). At the same time, it is now well known that agglomerations are not islands and that in the present global economy, most cities interact at least to a certain degree (Fingleton 2003; McCann and Acs 2011). The growing awareness in the theoretical spatial literature of the rise of urban

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networks and increasing geographical scope of agglomeration externalities is reflected in the introduction of new terms, such as ‘urban network externalities’ (Capello 2000), ‘regional externalities’ (Parr 2002), ‘externality fields’ (Phelps 1992), and ‘borrowed size’ (Phelps et al. 2001; Meijers and Burger 2016). In this regard, McCann and Acs (2011) in fact argue that currently, national and international urban network embeddedness is more important for urban productivity than urban size, while according to Coe et al. (2004) and Yeung (2009), urban and regional development is becoming a ‘globalizing’ phenomenon. Similarly, Florida (2008) perceives globally interconnected cities as the real engines of economic growth.

This discussion not only suggests that the embeddedness of cities in regional, national, and international networks is important for their performance but also that small and medium-sized cities could potentially internalize the benefits of larger cities by being well-positioned in urban networks. Unfortunately, relatively little is known about the relative importance of urban network externalities *vis-à-vis* agglomeration externalities or which types of cities would profit from urban network externalities. This special issue of *Papers of Regional Science* focuses on the rise of urban network externalities: external economies from which firms and households can benefit by being located in agglomerations that are well embedded or positioned in networks that connect with other agglomerations.

1.1 *The spatial and network dimension of external economies*

Paramount to the study of agglomeration externalities is the discussion of what constitutes a city or industrial agglomeration. Prior to the Industrial Revolution, it was relatively easy to define a city. As a general rule, a city could be identified as an agglomeration of people and businesses, separated from the countryside by fortifications, gates, and towers. This characterization is not only reflected in the Greek term *polis* (‘surrounding wall’) and the Latin word *urbs* (‘walled city’) but also in the English ‘town’ (‘circular palisade’), the German *zaun* (‘hedge’), and the Spanish *ciudad* (‘stronghold’) (Kerbat 1995). However, what primarily distinguished cities from rural areas in medieval times were their specific legal status (notably ‘city rights’) and their political identity. Medieval towns were usually autonomous, being free from feudal bondage and having their own jurisdiction (Hohenberg and Lees 1995). Perhaps this was the moment in history when the benefits of cities could truly be labelled ‘agglomeration economies’ because in spatial terms, they coincided perfectly with agglomerations and could not be enjoyed outside the city walls.

Since the Industrial Revolution, we have witnessed the increasing spatial disintegration of ‘agglomeration’ and the ‘benefits and costs’ known as ‘agglomeration economies and agglomerations diseconomies’. Indeed, cities have shifted their functional borders as mobility has steadily increased and communication technology has developed further. Trams, trains, and the telegraph in the nineteenth century and telephones, automobiles, planes, and the internet in the twentieth century are examples of technology that have affected the spatial extent of the city (Anas et al. 1998). Paralleling the ever-growing suburbanization in the early twentieth century, the focus on urban research gradually shifted from the internal structure of cities (‘city as a system’) to the external relations of cities, also referred to as ‘systems of cities’ or ‘urban systems’. The notion of urban systems would be formalized in the interbellum in central place theory (Christaller 1933; Lösch 1944), which would later gain popularity in the fields of economics and geography. Although Christaller’s and Lösch’s works predominantly focused on explaining the distribution, size and number of cities and towns and the location of economic activity, a common key element in their theories was that not all places can be self-sufficient and places are inherently interdependent.

The academic literature that developed after the Second World War that would build on the work of Christaller (1933) and Lösch (1944) followed two paths (McPherson 1981). On the one hand, a number of regional science and urban economic studies have extended and modified the

formal theoretical model to arrive at a more general and realistic model of urban systems (Berry et al. 1988). In particular, the regional science movement combined the work of Christaller and Lösch with insights about empirical regularities in spatial distributions. These insights included the rank-size rule and the distance-decay functions and gravity models as well as general equilibrium models. Contemporary work on the external relations of cities that builds on this literature includes urban economic models on polycentric cities and suburbanization (Fujita and Ogawa 1982) and the evolution of city-systems (Mori 1997; Fujita et al. 1999) as well as the complex cities and networks literature (e.g., Wilson 2000; Batty 2008; Reggiani and Nijkamp 2009; Roth et al. 2011).¹

On the other hand, empirical research, mostly originating from the urban systems school (see, e.g., Berry 1964; Pred 1977; Bourne and Simmons 1978), has empirically studied the organization of urban systems and the nature of urban networks. In this literature, urban systems have been generally defined as a set of regionally, nationally or globally linked and interdependent urban areas (Bourne and Simmons 1978). To date, this strand of research can be found in the literature on world cities and global production networks (e.g., Taylor 2001; Derudder et al. 2010; Jacobs et al. 2010; Rozenblat 2010; Wall and Van der Knaap 2011)² as well as the literature on urban networks and regional spatial structure (e.g., Camagni 1993; Camagni and Salone 1993; Batten 1995; Van Oort et al. 2010; Burger and Meijers 2012).³

Despite the attention that has been given to the external relations of cities over the past 70 years, the literature on agglomeration externalities in the late twentieth century that has built on the work of Marshall (1890) and Hoover (1948) predominantly focuses on the economics of physical proximity and not on the economics of the external relations of cities. The seminal work by Glaeser et al. (1992) and Henderson et al. (1995) that placed agglomeration externalities in the centre of the economic debate linked economic geography to the endogenous growth theory of Romer (1986) and Lucas (1988).⁴ The large empirical literature that has emerged focuses on how firms are influenced by co-location and which spatial conditions cause greater than proportional growth in productivity and economic activity. Most notably, one of the key debates in this literature has been whether diversity (urbanization economies) or specialization (localization economies) best promotes growth,⁵ where more recent contributions in the field of regional science and urban economics have also paid attention to firm heterogeneity (e.g., Henderson 2003; Baldwin et al. 2008; Ottaviano 2011; Stavropoulos and Skuras 2016) as well as examining the channels of sharing, matching and learning through which agglomeration externalities occur (Duranton and Puga 2004; Puga 2010).

At the same time, discussions of the geographical scale of agglomeration externalities and urban network externalities have been relatively scarce (Parr 2002; Johansson and Quigley 2003; Phelps and Ozawa 2003; Burger et al. 2009; Camagni et al. 2015). However, it is unfair to say that ideas on urban networks are completely absent in the literature on agglomeration externalities. Already in 1931, the British economist Robinson introduced the distinction between immobile external economies and mobile external economies. Focusing on the Marshallian-type of localization externalities, Robinson (1931) argued that immobile external economies depended on the size of an industry in a locality, while mobile external economies depended on the size of the industry in a country or the world as a whole and can be shared by firms outside the locality from which they arise. This sharing can take place through the existence of functional relationships between actors in

¹ For a re-appraisal of central place theory in the regional science literature, see Mulligan et al. (2012).

² See, also, recent special issues in *Global Networks* (Derudder and Witlox 2010), *Urban Studies* (Derudder et al. 2010), *Tijdschrift voor Economische en Sociale Geografie* (Derudder and Parnreiter 2014).

³ An overview of these different works can be found in Burger (2011). Recent special issues have appeared in *Regional Studies* (Hoyler et al. 2008; Burger et al. 2014).

⁴ See McCann and Van Oort (2009) and Neffke (2009) for excellent overviews of this literature.

⁵ For meta-analyses of this literature, see Melo et al. (2009) and De Groot et al. (2016).

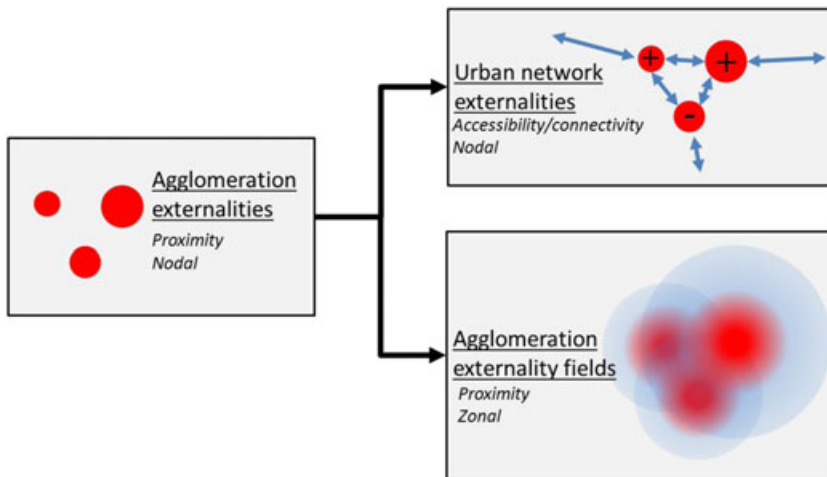


Fig. 1. Possible spatial transformations of agglomeration externalities

different localities. Key in the distinction between agglomeration (immobile) and urban network (mobile) externalities is that the former benefits are spatially constrained and diminish with distance (see also Van Meeteren et al. 2016), while the latter are not spatially constrained and attenuate with the strength of functional relationships between cities.⁶

In the 1970s, Alonso (1973, p. 200) would introduce the related concept of ‘borrowed size’, by which he meant that ‘a small city or metropolitan area exhibits some of the characteristics of a larger one if it is near other population concentrations’. More precisely, he suggested that smaller urban areas ‘borrow’ some of the agglomeration benefits of their larger neighbours, while avoiding their agglomeration costs. The phenomenon of ‘borrowed size’ demands a network perspective because accessibility and network connectivity to large city agglomeration externalities are prerequisites for borrowed size to occur. Although Alonso predominantly took a regional perspective, Meijers and Burger (2016) recently argued that the concept can be easily generalized to higher spatial scales.

Recognizing the importance of incorporating the network dimension in the study of external economies and urban growth, two strands of research are now gradually taking root. These strands of research both suggest that agglomeration externalities are spatially transforming into a sort of ‘agglomeration externality fields’, or into ‘urban network externalities’ (see Figure 1).⁷

1.2 Agglomeration externality fields

First, it is increasingly recognized that agglomeration benefits and, perhaps to a lesser extent, costs seem to be regionalizing, covering an ever larger territory beyond the agglomeration in

⁶ For a more general definition and discussion of ‘network externalities’, see the work by Katz and Shapiro (1985).

⁷ Parallel to these developments, the geography of innovation literature has examined the micro-foundations of knowledge spillovers and the importance of different kinds of proximity (geographical, institutional, organizational, social) for learning and innovation (cf. Boschma 2005). One key finding in this literature is that spillovers can take place over longer distances and not only geographical proximity matters for learning and innovation (e.g., LeSage et al. 2007; Ponds et al. 2010). However, given that this literature does not primarily focus on urban systems, it is not extensively discussed in this introduction to the special issue.

the strict sense of the word. When agglomeration externalities are no longer nodal, but zonal, it would make sense to refer to them as ‘externality fields’ (Phelps et al. 2001) or ‘regional externalities’ (Parr 2002), and some authors have suggested distinguishing the traditional agglomeration externalities from these ‘regional externalities’ (Parr 2002). Alternatively, several studies have examined the spatial extent of agglomeration externalities using – among others – spatial econometric techniques. In a recent overview provided by Drucker (2012), it was however noted that the empirical evidence obtained so far is quite inconsistent in that some studies find that agglomeration externalities rapidly decay with distance (e.g., Rosenthal and Strange 2003; Van Soest et al. 2006; Baldwin et al. 2008), while other studies conclude that agglomeration externalities may well extend over 100 kilometres (e.g., Partridge et al. 2007; Renski 2011). Most likely, distance decay depends on the industry, agglomeration type, and the type of externality that is examined (Drucker 2012). Further research is needed but is beyond the scope of this special issue.

1.3 Urban network externalities

Second, urban networks seem to provide a substitute for the benefits of agglomeration. In this regard, the performance of cities may become increasingly dependent on their position in urban networks that can have a national, continental or worldwide scope, where external economies become increasingly mobile. Capello (2000) introduced the term ‘urban network externalities’,⁸ referring to the benefits originating from the functional relationships between places, where the presence of functional networks between cities could create synergies and complementarities (see also Camagni 1993; Boix and Trullén 2007; Van Oort et al. 2010). Related to the discussion on urban network externalities has been the re-introduction of Alonso’s concept of borrowed size (see Phelps et al. 2001; Burger et al. 2015; Meijers and Burger 2016). Borrowed size processes take place when a city displays performance levels normally associated with larger cities. This is enabled through interactions in networks of cities across multiple spatial scales that serve as a substitute for the benefits of agglomeration (Meijers and Burger 2016). Accordingly, small and medium-sized cities could potentially internalize the benefits of larger cities by being well-positioned in urban networks. At the same time, it should be noted that network connectivity can also have a negative effect on the performance of cities due to competition processes and not every city profits equally from high levels of network embeddedness.

2 Contents of the special issue

This special issue brings together a multidisciplinary team of authors with backgrounds in urban economics, regional science, economic geography, and urban planning, providing different perspectives and approaches in the study of networks and agglomerations. Broadly speaking, the papers address three issues. First, the term ‘urban networks’ suffers from a lack of specificity, and hence, the conceptualization of urban network externalities is not always clear. Second, some provide evidence on the extent to which urban network externalities influence urban performance and how they compare to traditional agglomeration externalities. Third, the heterogeneous relationship between urban network externalities and urban performance is addressed because not all places might profit from network connectivity. Several of the papers address multiple issues. Their main contributions to the debate on the rise of urban network externalities are highlighted below.

⁸ The terms ‘urban network externalities’ (Capello 2000) and ‘city network externalities’ (e.g., Meijers et al. 2016) can be used interchangeably.

Glaeser et al. (2016) build a model of endogenous amenities and recombinant growth to answer the urgent policy question of how to accommodate future urbanization: Would it be more beneficial to develop urban networks (groups of cities enabling the exchange of goods, people and ideas) or foster the rise of larger or denser mega-cities? Their theoretical model suggests that the answer to this question depends on the trade-off between local returns to scale in innovation, housing supply elasticity, and the importance of local amenities. Furthermore, historical development pathways and place-specific conditions need to be taken into account.

Van Meeteren et al. (2016) call for prudence in the debate on agglomeration and networks, as both are fuzzy concepts and need to be disentangled before one can arrive at conclusions regarding their significance in explaining the performance of cities and regions. This disentangling is less straightforward than it may seem as the concepts of agglomeration and network are inter-related. Their conceptual paper provides a review of the concepts of agglomeration externalities and urban network externalities and presents a combinatorial typology of agglomeration and networks derived from three disciplinary perspectives (industrial organization, spatial-economic, and geometrical). Subsequently, they demonstrate how their conceptual approach can be adopted in empirical research on globalized producer services firms.

One type of network is made up of international supply chains. Fujita and Hamaguchi (2016) emphasize the benefits of such networks in terms of increasing the efficiency of human resources, productivity and welfare, and the potential for inclusive globalization. However, longer and more complex networks come with a cost, as uncertainties and risks increase. Volatility is intrinsic to international supply chains. Fujita and Hamaguchi document the process of supply chain internationalization in East Asia, specifically focusing on its spatial implications, and show how idiosyncratic supply shocks triggered by natural disasters affected trade patterns.

The paper by Duranton (2016) explains city growth in Colombia. He includes a more comprehensive set of determinants than previous research, providing an elaborate review of a wide variety of factors that are essential to understand urban growth, and simultaneously sketches how these factors can best be empirically explored from the modelling point of view. A thorough empirical analysis of these determinants is provided. Road connectivity and market access are examples of factors associated with urban network externalities. Road connectivity in Colombia primarily facilitates travel within cities and fosters growth, as it has done for more than a century. However, somewhat mixed and scarcely significant results are found for the importance of being able to access other cities and their market potential in the Colombian context.

The paper by Camagni et al. (2016) questions the too simplistic reasoning that further agglomeration leads to growth, suggesting that further urbanization of large mega cities is not the key to welfare increases, especially not in recent years. While larger cities have higher productivity, urban growth does not necessarily imply increases in productivity. They call for such a dynamic perspective, which demands: (i) a focus on net agglomeration benefits rather than gross benefits as is typical for many studies; (ii) breaking down the factor 'size' into multiple determinants, including network embeddedness; and (iii) the explicit introduction of the time dimension in theory and empirical analysis. They find that large and small cities have the same opportunity to grow, which suggests that other factors than size drive urban growth. These other factors include among others the increase in the presence of urban functions – in the city itself but also in nearby places ('borrowed functions').

Thissen et al. (2016) decompose the growth of European regions into demand-led growth due to growing export markets and structural growth due to growing export market shares in those export markets. The vast majority (80%) of regional growth (and possibly decline) is explained by demand-led growth in export markets. As structural growth is mostly affected by regionally varying characteristics, it can be inferred that these characteristics – including regional policies – are of limited importance for the growth opportunities of regions. At the same time, the authors show that regions operating on similar markets are affected by the same

type of externalities because they operate on the same markets in a trade network. In this regard, these results suggest that recently advocated place-based development strategies of European regions should be complemented with network strategies.

Finally, the study by Meijers et al. (2016) argues that urban network externalities make it possible to ‘borrow size’ in order to explain the performance of small and medium-sized cities in Western Europe. They establish that the importance of agglomeration (size) and network connectivity differs across metropolitan functions. Sometimes network connectivity is more important than size in explaining the presence of certain metropolitan functions in cities, suggesting that networks can indeed substitute for size. It is also shown that small cities profit more from an increase in size, whereas larger cities profit more from an increase in network connectivity.

3 Towards a research agenda

Based on a synthesis of contributions to this special issue and the existing literature, the contours of a research agenda on the rise of urban network externalities become apparent. To advance the debate on agglomeration and networks, we identify five urgent research issues related to the conceptualization of network externalities, the trade-off between agglomeration and urban network externalities and the heterogeneous impact of urban network externalities:

1. multiplexity and heterogeneity in networks and their impacts;
2. interrelations between agglomerations and networks;
3. a dynamic perspective on agglomeration and networks;
4. a place-based perspective on agglomeration and networks; and
5. policy implications of a focus on networks.

3.1 *Multiplexity and heterogeneity in networks and their impacts*

Camagni et al (2016) show that small cities manage to profit relatively more from a developed network of co-operative relationships with other cities than do large cities. Meijers et al. (2016), using a rather different measure of networks, arrive at the opposite, finding that smaller cities are less able to profit from their networks. Likewise, larger cities seem to be more dependent on international networks, whereas smaller cities perhaps benefit more from regional networks (Burger et al. 2015; Glaeser et al. 2016; Meijers et al. 2016). This issue calls for future research to address multiplexity and heterogeneity: The effects of different types of networks (multiplexity) may be different for different types (sizes) of cities (heterogeneity). Glaeser et al. (2016) identify another type of heterogeneity that requires exploration, suggesting that the balance between the importance of networks and agglomeration varies from region to region, which depends, for instance, on the urban structure. In addition, Thissen et al. (2016) show that connectivity is important for some sectors, but certainly not all. In sum, different types of networks will have different impacts on cities and firms or may substitute for some agglomeration externalities but certainly not all. Further research is required to disentangle different types of networks and their impact, which may vary according to the city and firm.

3.2 *Interrelations between agglomerations and networks*

In addition, the more general issue is how agglomeration and networks are interrelated and how they can be meaningfully disentangled despite their interrelatedness, both conceptually and in empirical frameworks. The urgency and relevance of this question is highlighted in

Van Meeteren et al. (2016), who draw attention to the causalities between the development of agglomerations and networks between them and the change over time of this relationship. However, this relationship is also crucial in supply chain internationalization, as regional development depends largely on the ability to couple specific local production assets with the global production networking strategies of firms, as Fujita and Hamaguchi (2016) mention. In addition, investment in particular agglomeration-related regional assets can only pay off when this leads to an enhanced market position in a distant geographical market as a result of competitive trade relations (Thissen et al. 2016). Duranton (2016) addresses a potential estimation bias when he suggests that urban growth and other urban outcomes may cause rather than follow network embeddedness, while these outcomes are also determined by many factors related to agglomeration. Camagni et al (2016) suggest that while size is often used as an explanatory variable of urban efficiency because it captures many factors correlated with size, this may not be correct as some of these factors, in particular the capacity to establish and maintain networks with close-by and distant cities, are increasingly less correlated with size. Thus, we need a better understanding of the relationship between agglomeration and networks.

3.3 A dynamic perspective on agglomeration and networks

Glaeser et al (2016) also rightly call attention to the time dimension, suggesting that urban networks may perhaps be more efficient at present in Europe, but this may in the long run lead to less growth due to a lower level of new firm creation. The long-run benefits of agglomeration and networks may be quite different from the short-run benefits. That such a dynamic perspective is enlightening is further demonstrated by Camagni et al. 2016, and demands attention.

3.4 A place-based perspective on agglomeration and networks

The study by Duranton (2016) warns us that the typical determinants of urban growth in the developed world are not necessarily the same in the developing world, even in a quite advanced developing nation as Colombia. This also holds for the determinants for network embeddedness that take central stage in this special issue. The importance of networks found in Meijers et al. (2016) and Thissen et al. (2016) for Europe is not demonstrated in the Colombian context. Similarly, Glaeser et al. (2016) state that whether a focus on developing urban networks or megacities is preferable is place-dependent. The mismatch between agglomeration theory and urban dynamics specifically in Europe (except Eastern Europe) is documented in Camagni et al. (2016) and Meijers et al (2016). More generally, this calls for prudence in generalizing research findings and incorporating place-specific contextual determinants in the analysis. The importance of local factors in mediating the role of urban networks needs to be further explored.

3.5 Policy implications of a focus on networks

In the future, attention should also be devoted to the policy implications of a rise in urban network externalities. Some of the findings presented in this special issue suggest a quite drastic reorientation of policies. Thissen et al. (2016) indicate that both location and network-specific policies are needed, while the current focus is predominantly on place-based development (e.g., ‘smart specialization’). Such network policies can focus on enhancing the position of a region or place in a wide variety of networks, ranging from networks fostering broad connectivity to dedicated networks of knowledge exchange, etc. Given the potential substitution effects of networks, it would seem that huge investments in large (capital) regions to stimulate further urbanization of large cities and hence agglomeration benefits are not necessarily more beneficial in fostering productivity than stimulating network development of these and other (also smaller) cities. Such

issues are addressed in a theoretical mode by Glaeser et al. (2016), but they emphasize that empirical estimations of their theoretical model are needed before final conclusions can be drawn on the trade-off between urban networks and megacities. Doing so involves solving a number of methodological issues related to the connection between urban size and amenities, in particular the endogeneity of size, and requires a better understanding of housing supply elasticities and evidence on returns to scale in idea creation. Stimulating integration between nearby places to strengthen regional urban systems seems a no-regret option and is particularly beneficial for medium-sized or 'second' cities (Cardoso and Meijers 2016). However, as Fujita and Hamaguchi (2016) show, there are also negative network externalities, and policies should also try to reduce them. However, the extent to which negative network externalities outweigh the positive ones should also be addressed in future research.

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Resumen. Este artículo presenta el tema de la edición especial ‘Aglomeraciones y la aparición de las externalidades de red urbanas’. Las externalidades de red urbanas se definen como las economías externas de las que se pueden beneficiar las empresas y los hogares situados en aglomeraciones que están bien arraigadas en redes que conectan con otras aglomeraciones. Las contribuciones se centran en la conceptualización de las externalidades de red urbana y su influencia en el desempeño urbano. Por último, se presenta un programa de investigación, que debería centrarse en: la multiplicidad y heterogeneidad de las redes y sus impactos; las interrelaciones entre las aglomeraciones y las redes, y su naturaleza dinámica y basada en la localización; y las implicaciones políticas de las externalidades de red urbanas.

要約: 本論文では、「集積化と都市ネットワーク外部性の出現」という特別な問題をテーマとする。都市ネットワーク外部性とは、企業または家計が、他の集積地と集積をつなぐネットワークに十分組み込まれた集積地に立地することにより、便益を享受できる外部経済と定義される。本論文では、都市ネットワーク外部性の概念化とその都市パフォーマンスに対する影響に注目する。最後に、ネットワークの多重性と多様性およびその影響、集積とネットワークおよびそのダイナミックで地域固有的な特徴との相互関係、都市ネットワーク外部性の政策的インプリケーションに焦点を当てるべきとする研究課題を提示する。